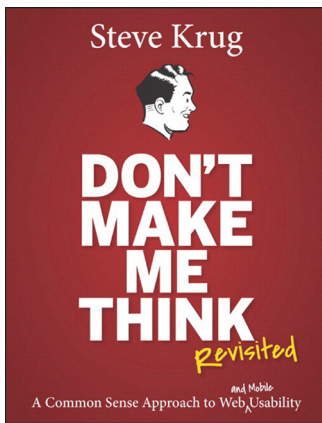




DON'T MAKE ME THINK, REVISITED

A Common Sense Approach to Web Usability

STEVE KRUG



The Big "So What"

Today, it's hard to imagine any business without a website or internet presence. **This book offers valuable principles and guidelines on how to design great, usable websites. Such insights are relevant for web designers, developers, and anyone who wants to think like a usability expert—to understand how people behave and how you can make things more usable.**



Introduction



This book was first written in 2000. Since then, technology has changed rapidly, yet the principles in the book remain unchanged because usability is fundamentally about human psychology. **Once you understand how the human brain works, you can continue to apply the insights even as technology and landscapes evolve.**



This 2013 edition was updated with newer examples and to address 3 additional landscape changes since 2000:

- Internet-usage has gone mobile. We no longer surf the web only in front of the computer. We're now constantly connected to the internet via different devices (especially our phones), to make dinner reservations, adjust our home lighting or temperature, do our banking etc.
- Websites are becoming smarter and more powerful and can learn/anticipate what users need. In fact, users often get irritated when sites don't do *exactly* what they expect them to do.

KEY QUOTES

"Usability is about people and how they understand and use things, not about technology ...while technology often changes quickly, people change very slowly."

"The human brain's capacity doesn't change from one year to the next, so the insights from studying human behavior have a very long shelf life. What was difficult for users twenty years ago continues to be difficult today."

—Jakob Nielsen

- The concepts of usability and user experience design (UX) are now widely known, not just to the techies. However, most people still don't know exactly what to do about it.

 As a usability consultant, the author helps to test websites to identify key areas for improvement. However, not every company can afford to hire consultants. This book equips you and your team with useful principles and tips to prevent and address usability issues on your own. In this summary, we'll use the term "website" to refer generically to both websites and web/mobile apps.

Key Principles and Guidelines

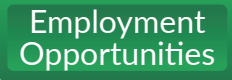
The 3 Laws of Usability

Basically, **something is usable** if an average person can figure out how to use it to achieve an outcome without it being more trouble than it's worth. Krug breaks this down into 3 laws:

 **Law #1: Don't make the user think.** This is the overarching rule. Each time a user has to pause (even for a split-second) to think about something, it distracts him from the action you want him to take. The goal is to make your website *effortless* to use, i.e. make it self-explanatory, if not self-evident.

- A website is self-evident when users "get" what it's about and how to use it without having to think. As a rule of thumb:
 - (i) Make clickable links/buttons obvious.
 - (ii) Use words that are obvious to everyone. Avoid technical jargon, clever-but-confusing marketing phrases, or terms that're specific to your industry/company.

Here are 2 examples:

	Obvious	Requires some thought	Unclear
#1			
#2			

KEY QUOTES

"Usability is, at its heart, a user advocate job...Usability is about serving people better by building better products."

"Your job is to get rid of the question marks...every question mark adds to our cognitive workload, distracting our attention from the task at hand."

"Making every page or screen self-evident is like having good lighting in a store: it just makes everything seem better."

- Even if you're doing something new that's unfamiliar to most people, you can still make the UX self-explanatory. Help users to understand and navigate the website with just a small amount of carefully-worded texts.



Law #2: Make every click an obvious choice with no need to think. Krug believes that the number of clicks doesn't matter, so long as each click is (i) mindless (i.e. no thinking required) and (ii) obvious (i.e. the user is sure it's the right choice). The only exception is during slow internet speed, in which case the # clicks will make a difference.



Law #3: Half the words on each page, then half them again. Remove all unnecessary words to reduce distractions, allow the key content to stand out, and shorten the page to minimize scrolling. The only exception is for news or content-driven articles.



How People Use the Web

We do *not* read websites in a sequential, detailed or orderly fashion. Instead, we scan webpages and navigate sites using trial-and-error.



We scan (not read) webpages. We almost never read a webpage from top to bottom.

- We tend to skim through a page quickly and zoom in on words that (i) are clickable and (ii) seem relevant, i.e. they match our immediate task/goal, personal interests or fit certain pre-programmed triggers (e.g. words like "Free", "Sex" or "Sale").
- When something looks roughly like what we're seeking, we'll click it, scan the new page, then go "Back" if it's the wrong choice. When we find a relevant page, we also scan it quickly to pick up the specific bits of info we need.



We make reasonable (not optimal) choices. We don't take the time to compare options and select the best one. We "satisfice" or click on the first reasonable option we see. After all, we can always go "Back" if it's not what we're looking for.

KEY QUOTES

"If you can't make something self-evident, you at least need to make it self-explanatory."

"It doesn't matter how many times I have to click, as long as each click is a mindless, unambiguous choice."

"We don't read pages. We scan them."

"We don't make optimal choices. We satisfice."



We go for guesswork (not the “right” approach). Instead of reading instructions or figuring out the details, we make assumptions and fumble our way through using trial-and-error. So long as we eventually find what we want, we don’t care if the process is inefficient. And, if something works, we stick to it until we stumble upon a better alternative, i.e. we won’t deliberately search for better options. If you can help users to “get” your site and achieve their goals faster, they’re more likely to return. You’ll also have a higher chance of (i) directing them where you want them to go, and (ii) showing them more benefits and features that you offer (not just what they stumble upon).



Design your Website for Easy Scanning

Help users to quickly understand your site and find what they need.



Make use of existing conventions i.e. common standards, designs or layouts. For example, a car’s acceleration pedal is *always* to the right of the brake pedal. The STOP sign—with its distinctive octagon shape and red color—is also universally understood.

- Similarly, there’re conventions for webpages, such as (i) where things are located (e.g. logos on the top-left corner, key navigation menu on the top or left), and (ii) familiar icons (e.g. shopping carts for ecommerce, magnifying glass for search bars).
- Conventions are widely-adopted because they work. Don’t break from convention just to be creative. Follow conventions unless your alternative is (i) just as clear/intuitive to the users or (ii) adds so much value that it’s worth the extra thinking/learning you’re putting them through.



Use effective visual hierarchies. Clearly show the relationship between various elements on the page, i.e. which things are important, similar to, or are part of other things. We encounter such visual hierarchies every day, e.g. in magazine or newspapers headlines and columns.

KEY QUOTES

“We don’t figure out how things work. We muddle through.”

“For most of us, it doesn’t matter to us whether we understand how things work, as long as we can use them. It’s not for lack of intelligence, but for lack of caring.”

“Innovate when you know you have a better idea, but take advantage of conventions when you don’t.”

- Make important items more prominent, e.g. bigger, more colorful, higher on the page and/or with stronger contrast.

Very important

Less important

Not that important

- Connect similar/related things visually, e.g. group similar content under a common heading, place them in a clearly-defined section or use the same design style/color.

Set A
Set B
Set C

Item 1
Item 2
Item 3

- Show that something is a sub-set of another by inserting it directly within the parent section, e.g. place all strategy-related books under a “Strategy” section.

Strategy Books

Book #1
Etc. etc. etc.

Book #2
Etc. etc. etc.



Format your content for easy scanning.

- Divide each page into clearly-defined sub-sections so users can quickly zoom in on the most relevant areas.
- Use formatting, such as different header levels (with corresponding font sizes), short paragraphs, bulleted lists and highlight the key words/phrases.



Make every click mindless and obvious (Law #2):

- Make clickable elements evident. Users will scan the page for what’s clickable, e.g. buttons, menu bars, words with specific colors and underlining. Don’t confuse them by using the same colored font for clickable and non-clickable headings.
- Use unambiguous choices. “Single-user or Multiple-users” is unambiguous but “Home or Office” isn’t. If an unambiguous choice isn’t possible, then guide users with form/tool tips that are (i) concise (with minimal info), (ii) timely (offered just when they need it), and (iii) unavoidable (i.e. they have to notice it).



Remove distractions or “noise” on your website. These include:

- “Loud” pages where everything shouts for the users’ attention. If a page is full of exclamation marks, bright colors, fancy fonts types, animations and popups, the user won’t know what to focus on.

KEY QUOTES

“The more important something is, the more prominent it is.”

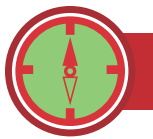
“Almost anything that can be a bulleted list probably should be.”

“Shouting is usually the result of a failure to make tough decisions about which elements are really the most important and then create a visual hierarchy that guides users to them first.”

- Messy pages that're so untidy and disorganized that it's hard to read or navigate.
- Clutter from unnecessary content which make it hard to find the useful stuff. Remove everything that doesn't really contribute to your page, including unnecessary words from:
 - (i) Social talk or welcome/intro messages that don't convey anything useful.
 - (ii) Lengthy instructions. If absolutely necessary, insert concise instructions only at the point where users will need them.



Navigation, Home Page and Testing



Navigation



Navigating a website is like looking for something in a huge departmental store.

- There're a few similarities:
 - (i) People are usually looking for something specific;
 - (ii) Some prefer to browse on their own first while others prefer to ask for help; and
 - (iii) They'll leave if they can't find what they're looking for.
- However, there're also differences. Unlike in a physical store:
 - (i) You can't tell how much of the website is unexplored;
 - (ii) You don't have a physical sense of where you are on the site; and
 - (iii) It's harder to return to a specific place in a website. Most people use the "Back" button, bookmarks (or stored personal shortcuts) and the Home Page (as a "base" to start afresh).

KEY QUOTES

"In the face of limited time and attention, everything that's not part of the solution must go."

"Happy talk is like small talk... But most Web users don't have time for small talk; they want to get right to the point."

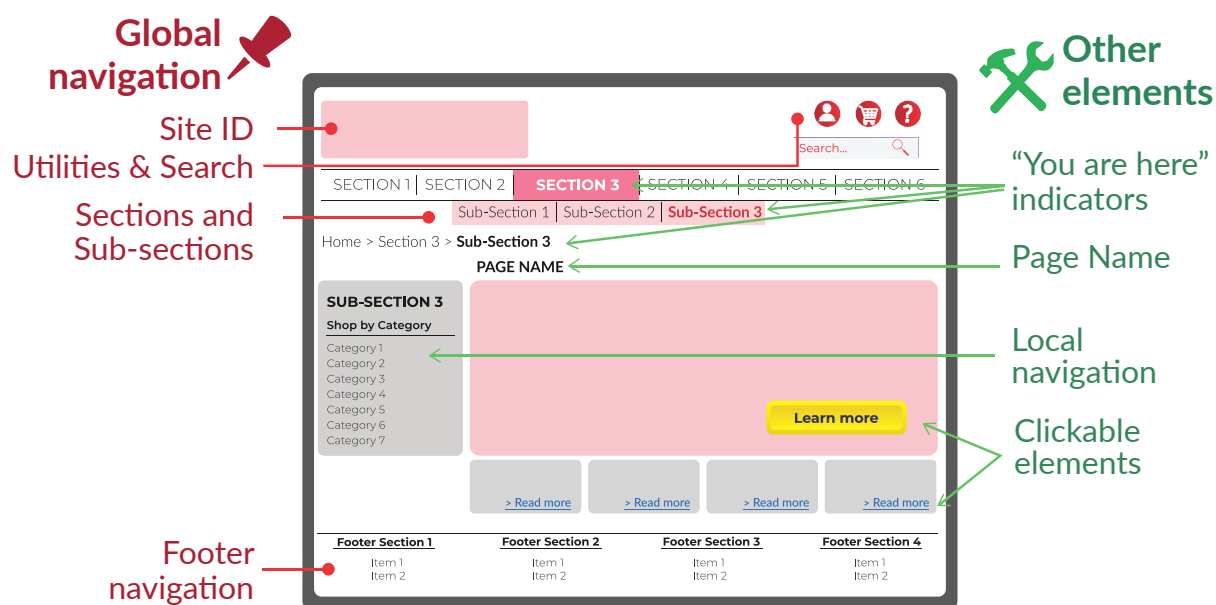
"It's a fact: People won't use your Web site if they can't find their way around it."



Web navigation serves 2 key goals: help users (i) find what they're looking for and (ii) know where they are on the site. Great navigation makes the site hierarchy clear, helps people to see what the site contains, how to use it (e.g. where to start and what options are available), and builds confidence in your brand.



We expect to find street signs at street junctions and corners, index pages in books, signs along supermarket aisles, etc. We feel frustrated when these expectations aren't met. Likewise, there are certain conventions for web navigation that users expect:



Global navigation (or persistent navigation) means having core navigation elements on every page of a site to provide a consistent experience. The only exception is when the user is filling up a form (e.g. to register or make a purchase).

- The Site ID or logo must appear consistently on every page so users know they're on your site. The Site ID usually doubles up as a Home page link, but you can still add an explicit link to "Home".
- Sections (or primary navigation) provide links to the key parts of your site. You can also include sub-sections (or secondary navigation) under each section.
- Utilities are links to other important webpages that aren't part of the main content hierarchy, e.g. login, shopping cart, search bar, about us, FAQs. Make them less prominent than the sections, include no more than 5 most-used utilities in the global navigation and put the rest in the footer.

KEY QUOTES

"Clear, well-thought-out navigation is one of the best opportunities a site has to create a good impression."

"It's vital to have sample pages that show the navigation for all the potential levels of the site before you start arguing about the color scheme."

- Include a search bar or minimally provide a link to a search page. Keep it straightforward, e.g. label it “Search” (not “Find” or “Quick Search”). Only split the search further (e.g. “Search by authors” or “Search by title”) if absolutely necessary.
- Most sites only display up to 2 levels of their content hierarchy. It’s much better to work out and show all levels of the hierarchy.



Always have a page name for every page. This should (i) be prominent, (ii) match the link that was clicked (to minimize confusion), and (iii) be located such that it frames the page content.

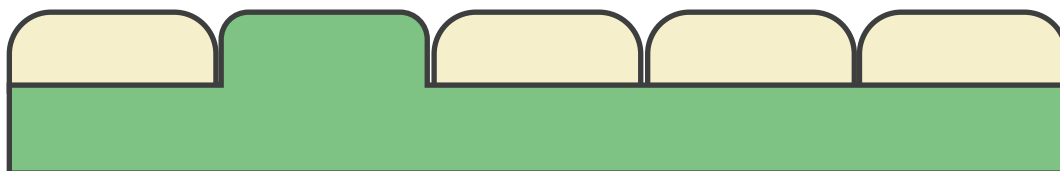


Let users know “you are here” on the site by:

- Prominently highlighting the sections/sub-sections that they’re at (e.g. using a pointer, bolding the text, using a different color for the text or button)
- Using breadcrumbs at the top of the page to tell users where they are relative to the Home page (e.g. Home > Store > Men’s Clothes > **Pants**).



Tabs are easy to understand and can be designed to be prominent yet elegant. Make sure you use a strong color contrast and create a clear visual link between the tab and main content.



Since people can enter your website from any page (not just the Home page), you should test the navigation for your entire site:

- Imagine you’ve been blindfolded and suddenly open your eyes to face a random page on your website. You must be

KEY QUOTES

“Just as every corner should have a street sign, every page should have a name.”

“Designers love subtle cues, because subtlety is one of the traits of sophisticated design. But Web users are generally in such a hurry that they routinely miss subtle cues.”

able to instantly identify (i) what website this is (site ID), (ii) what page you're on (page name), (iii) what the key sections are (sections), (iv) what you can do at this level (local navigation), (v) where you are in the site, and (vi) how to search for what you need.

- Print out one of your pages and see if others can identify these elements at a glance: (i) site ID, (ii) page name, (iii) sections, (iv) local navigation, (v) "you are here", and (vi) search options.



Home Page



The Home page is one of the most challenging pages to design because you must fit in many things:

- It typically includes elements like your site ID and mission, site hierarchy, search functions, teasers/highlights (of key content, features and deals), shortcuts to commonly-used content/features, and registration or login forms.
- They must also meet different users' objectives by helping them to find what they're looking for, interest them with your offers, build trust/credibility, and show them where to start.



Many stakeholders will want to be featured on the Home page, but you mustn't clutter the page. Instead, **the Home page must give a clear, big-picture overview of your site.**

- Specifically, it must quickly answering these 4 questions:
 - (i) What is this site? [Site ID and mission]
 - (ii) What can I find here? [Content]
 - (iii) What can I do here? [Features]
 - (iv) Why should I be here? [Value proposition]
- The initial impression formed within the few seconds will affect how the user interprets (or misinterprets) everything else on that page.

KEY QUOTES

"Preserving the Home page from promotional overload requires constant vigilance."

"We make snap judgments, but they tend to be a pretty reliable predictor of our more reasoned assessments."

- Most people will also return to the Home page to orientate themselves—what they see will affect if/how they use your site.



These 3 elements can be used to highlight the key message upfront:

- A tagline is normally found with your logo and sums up your value proposition. Great taglines are short (<6-8 words), clear and explain specifically what you do, e.g. “wheels when you want them” (Zipcar). An ideal tagline captures your differentiating factor and core benefit.
- A welcome blurb is a succinct description usually located on the top left of the content space e.g. “Founded in 1996, Idealab is the longest running technology incubator. We have created over 150 companies with more than 45 IPOs and acquisitions.” Never use your corporate mission as the welcome blurb.
- A “learn more” button or link.

It’s ok to use more space to convey your key proposition, so long as the content is clear and succinct. Use pictures/videos and highlight only the most compelling content/features as teasers.



Once you’ve addressed the 4 questions, you must show the user where to start, e.g. to search, browse, get free samples, enjoy your promotions, or begin with Step 1 of a specific process. Ensure your entry points are distinct, e.g. your sections must look like sections, and your search bars must look like search bars. Label them clearly, e.g. “Browse by Category”, “Search”, “Sign in” or “Start free trial”.



Test every element on your Home page. Do not rely on intuition or assumptions.

- Web teams often waste time/energy debating over design elements, e.g. whether to use pull-down bars or a certain layout. Every stakeholder (from the designer to the CEO) has personal preferences about web usage. We tend to assume that others share our preferences, and we argue over what the “average user” wants. In reality, web usage

KEY QUOTES

“If people are lost when they start out, they usually just keep getting...lost.”

“Testing reminds you that not everyone thinks the way you do, knows what you know, and uses the Web the way you do.”

“What works is good, integrated design that fills a need—carefully thought out, well executed, and tested.”

is quirky and unpredictable. The best approach is to plan carefully then test if/how people actually use the elements.

- People also tend to test the wrong things, e.g. they test the design of a promo banner, only to realize that users aren't interested in the promotion at all.



How to do Usability Testing



What is usability testing?

- Usability tests ≠ focus groups. Focus groups are small groups of people who share their views/opinions about something. They may be useful in the *early* stages of planning to get ideas and feedback on your value proposition. On the other hand, usability tests are about watching how people use something (e.g. your website); they should be used at *all* stages of development, from prototype-testing to identifying/fixing specific problems.
- Any testing is better than none. A test earlier in a project is also many times more valuable than a test at the end of the project, since you'd have more time to use and apply the insights.



Professional testing can cost anything between \$5-10k. However, you can do DIY testing for a fraction of the cost.

- **Get your Web development team to spend just 1 morning a month to do usability testing:** do a test with 3 users, debrief over lunch, and decide what to fix before the next monthly test. The idea is to keep it simple so you have a fixed testing schedule that you're committed to. Agile development teams should do more frequent testing, e.g. 2x a month.
- Krug recommends a small sample size of just 3 users for each testing. After all, you don't need to uncover all the problems or to prove something statistically. **Your goal is simply to get actionable insights to improve your site every month.**
- You can use anyone for the tests. In most cases, the problems faced by 1 user will apply to other users, so there's no need for participants to match your user profile exactly. Usually,

KEY QUOTES

"People often test to decide which color drapes are best, only to learn that they forgot to put windows in the room."

"Testing always works, and even the worst test with the wrong user will show you important things you can do to improve your site."

"You can find more problems in half a day than you can fix in a month."

you can recruit participants (from Craigslist, social media, website popups etc.) for an incentive of about \$50-100 for 1 hour.

- Involve as many observers as possible. Put them in a separate observation room where they can see the participant’s screen and hear what’s happening in the testing room. After each session, take a short break and get each observer to write down the top 3 usability problems they observed.
- If you’re redesigning your website, start by testing what currently works or doesn’t work before you make the changes. You can also get participants to test a competitor site for additional insights.
- Each month, select the specific tasks and sub-tasks you want the participants to do. For example, to test account-logins, they may need to create an account, change their password, retrieve a lost password etc. You can also instruct them with specific tasks like “Find a summary for a book you read recently” or “Find a strategy book for less than \$15”.
- Here’s a sample test session structure. You can also consider alternatives such as (i) remote-testing (via screen-sharing) or (ii) unmoderated remote-testing (i.e. send people the task/links, then get them to record what they do).

Item	Description
Welcome (4 min)	Explain what to expect.
Questions (2 min)	Find out more about the participant
Home page (3 min)	Open your Home page and ask the participant for inputs. Assess if your page is easy to understand.
Tasks (35 min)	Observe the participants as they perform the assigned task(s). The facilitator can ask questions like “What are you thinking?” or “What are you doing now?”. Don’t say/do anything to influence guide them. If they ask for help, ask “What would you do if I wasn’t here?”
Probe (5 min)	Ask questions to clarify what happened, and/or present questions from the observers.
Wrap up (5 min)	Thank the participant and pay them.

KEY QUOTES

- Do the debrief over lunch.
- (i) List the issues on a whiteboard: Get everyone to share the top 3 problems they observed over the 3 sessions. Agree on the 10 most serious problems and rate each one on a scale of 1-10.
- (ii) List the top 10 issues in order of seriousness. Discuss how you can address each one, including who to fix it and how. Start from the most serious item and keep going until you've enough items for the available time/resources that month. Focus on the most serious items, but also include low-lying fruits can be easily fixed within 1 hour.
- Useful tips: Often, the solution is to remove/simplify an existing item, *not* add more stuff. If a user goes off-track but corrects himself without help or obvious frustration, the issue can be left alone. There's no need to fix a problem 100%; *any* improvement helps.



Other Considerations



Mobile Usability



Mobile growth exploded after Apple made it possible to scroll by swiping and to zoom in/out by “pinching” the screen. The principles for usability are generally similar for mobile and web. However, (i) you need to cater for a much smaller screen and (ii) there're fewer conventions since mobile technology is still evolving rapidly.



Fundamentally, mobile space constraints mean that you must make tradeoffs, i.e. decide what to leave out.

- One way is to design for mobile users first so you're forced to choose only the most vital content/features, before adding more content/features for the desktop version.
- You must still include all critical elements for the mobile version, but break up content into sub-parts or levels and

KEY QUOTES

“Focus ruthlessly on fixing the most serious problems first.”

“Mobile devices meant cramped devices, squeezing Web pages the size of a sheet of paper into a screen the size of a postage stamp.”

“Most of the challenges in creating good mobile usability boil down to making good tradeoffs.”

prioritize them. People will have to tap/scroll more, but they're ok so long as each step is obvious and mindless.

 Here are some tips you can apply immediately:

- Start with these 3 low-lying fruits: (i) allow zooming on your website (so users can still view elements that are too small), (ii) ensure external links (emails, social media etc.) bring people directly to relevant parts of the site (not the Home page) and (iii) give the option to view the full website instead of the mobile version.
- Make each element extremely clear, e.g. use 3D buttons that're obviously clickable, outline boxes where people should type in. If you're using Flat design, you must deliberately create other forms of differentiation. If you use "hovering" in your Web design (e.g. drop-down menus that appear when the cursor is above a section), remember to find alternatives for mobile users since hovering doesn't work on touchscreens.
- Speed: make sure your pages load quickly.

Basically, users must easily figure out how to use your app without complex tutorials. Once they "get" it, they should easily remember or relearn it when they return. You can delight your customers with surprising features, but never do it at the expense of usability.

 Usability-testing on mobile is similar to that on the web except for several logistical challenges, since the tools for web-testing are still not widely available on mobile. For example, it's harder for the facilitator/observers to see the user screen, and mobile screen-sharing is not widely available yet. Krug recommends:

- Pointing a camera at the mobile screen so you can capture both the screen and the users' fingers; and
- Attach the camera to the device so the user can hold it freely.



Other Factors and Influences



Every user has a finite pool of goodwill toward your brand. Each time you anticipate the user's needs and make him feel

KEY QUOTES

"Almost everything that's different when you're doing mobile testing isn't about the process; it's about logistics."

"There's another important component to usability: doing the right thing—being considerate of the user."

at ease, understood and competent, you increase the goodwill. Each time the user feels confused, unsure or faces an obstacle, the goodwill depletes.



Ideally, your website/app should be accessible for everyone, including those with disabilities. Even if you can't afford to develop customized solutions, you can start with 4 areas: (i) fix usability issues that affect everyone, (ii) include a few blind users in your usability-tests, (iii) read up on Web accessibility, and (iv) go for low-lying fruits, e.g. add alternate attributes for every image, use appropriate headings, use forms that work with screen readers, choose a web template that's designed to be accessible.



To convince your management to support usability improvements, you'd need to (i) show the returns on investment and (ii) present the benefits in their language (e.g. using their buzzwords or KPIs). A simpler way to start is to do a test on competitors (which is typically of interest to management), get the bosses to observe a usability test, or do your own test then share the results.

KEY QUOTES